PharmaSys

Pharmacy Management System

Developer: Davix HDM

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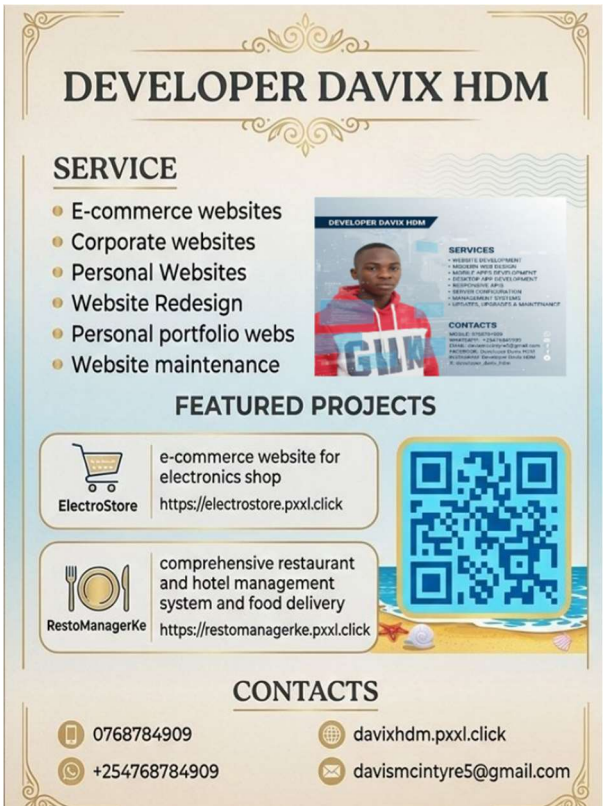

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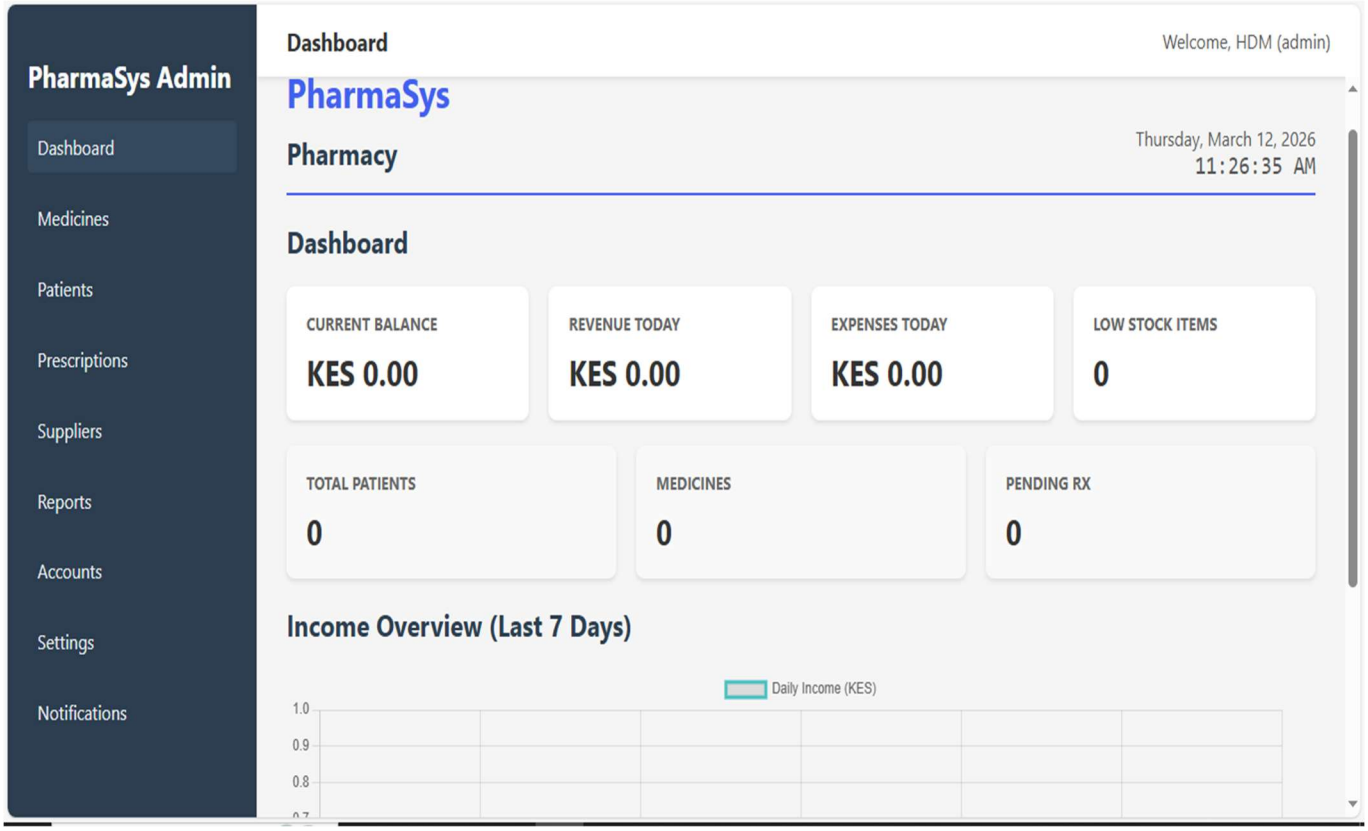
System Overview

Interface



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- **All beta testers** – who provided invaluable feedback during development.
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Introduction

PharmaSys is a comprehensive, full-stack pharmacy management system designed to streamline the daily operations of a modern pharmacy. Built with the MERN stack (MongoDB, Express.js, React, Node.js), it offers a robust, secure, and user-friendly interface for managing inventory, patients, prescriptions, financial accounts, and more. The system is deployed and accessible online, demonstrating its production readiness.

This document provides a detailed overview of the system's architecture, features, technology stack, API endpoints, deployment procedures, and future roadmap. It serves as both a technical reference for developers and a project summary for stakeholders.

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1. Features

- **Dashboard** – Real-time overview of key metrics: current balance, today's revenue/expenses, low stock alerts, patient counts, and a 7-day income chart.
- **Inventory Management** – Full CRUD for medicines, batch tracking, expiry alerts, and low stock notifications.
- **Patient Management** – Manage patient profiles, contact details, and prescription history.
- **Prescription Handling** – Upload, verify, and manage prescriptions with status tracking (pending, approved, dispensed, rejected).
- **Supplier Management** – Keep detailed records of medicine suppliers, including contact persons and addresses.
- **Financial Accounts** – Record income, expenses, and withdrawals; view current balance; print individual transaction receipts (with pharmacy name watermark).
- **Reporting Module** – Generate and print sales, inventory, and financial reports with custom date ranges; all printouts include a subtle pharmacy watermark.

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- **User & Settings Management** – Admin panel for managing system users (CRUD) and configuring pharmacy details, currency, and tax rate.
- **Role-Based Access Control** – Secure routes based on user roles: admin, pharmacist, cashier, manager.
- **Notifications** – In-app alerts for low stock, expiring medicines, and prescription updates.

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2. Technology Stack

Frontend (React)

- **React 18** – Functional components, hooks, and context API.
- **React Router v6** – Client-side routing with hash-based strategy for easy static hosting.
- **Redux Toolkit** – State management for user authentication and global UI state.
- **Formik & Yup** – Form handling and validation.
- **Chart.js & react-chartjs-2** – Interactive dashboard charts.
- **Axios** – HTTP client for API communication.
- **CSS3** – Fully custom responsive design (no UI framework).

Backend (Node.js + Express)

- **Node.js** – JavaScript runtime.
- **Express.js** – Web framework for building RESTful APIs.
- **MongoDB + Mongoose ODM** – Data modeling and persistence.
- **JWT (JSON Web Tokens)** – Authentication and authorization.
- **Bcrypt.js** – Password hashing.

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- **Multer** – File uploads (prescription images).
- **Node-cron** – Scheduled tasks (e.g., expiry alerts).
- **Nodemailer / Twilio** – Email/SMS notifications (optional, configured via environment).
- **Winston** – Logging.
- **CORS** – Configured for multiple origins (local and production).

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3. System Architecture

The system follows a classic client-server architecture with a clear separation between frontend and backend:

- **Client (React App):** Served as static files from a web server (e.g., Nginx). All dynamic data is fetched via API calls to the backend.
- **Server (Node.js/Express):** Exposes RESTful API endpoints, handles authentication, business logic, and database interactions.
- **Database (MongoDB):** Stores all application data in collections: users, patients, medicines, suppliers, prescriptions, transactions, settings, notifications.

Communication between client and server is stateless; each request includes a JWT token for authentication. The frontend uses hash-based routing, allowing it to be hosted on any static server without special configuration.

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4. Database Schema (MongoDB Collections)

users

- name (String)
- email (String, unique)
- password (String, hashed)
- role (String, enum: admin, pharmacist, cashier, manager)
- createdAt (Date)

patients

- name (String)
- phone (String)
- email (String)
- address (String)
- dateOfBirth (Date)
- allergies (Array)
- prescriptions (Array of ObjectIds referencing prescriptions)
- loyaltyPoints (Number)

medicines

- name (String)
- genericName (String)
- category (String)

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- manufacturer (String)
- price (Number)
- cost (Number)
- stock (Number)
- reorderLevel (Number)
- batches (Array of subdocuments: batchNo, expiryDate, quantity, supplier)
- requiresPrescription (Boolean)

suppliers

- name (String)
- contactPerson (String)
- phone (String)
- email (String)
- address (String)

prescriptions

- patientId (ObjectId)
- doctorName (String)
- issueDate (Date)
- expiryDate (Date)
- items (Array: medicineId, dosage, duration, quantity)
- status (String: pending, approved, dispensed, rejected)
- uploadedBy (ObjectId)
- imageUrl (String)

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accounttransactions

- type (String: income, expense, withdrawal)
- amount (Number)
- description (String)
- date (Date)
- createdBy (ObjectId)
- reference (String)

settings

- pharmacyName (String)
- address (String)
- phone (String)
- email (String)
- licenseNumber (String)
- currency (String, default: KES)
- taxRate (Number)

notifications

- userId (ObjectId)
- type (String: stock, expiry, prescription, general)
- message (String)
- read (Boolean)
- createdAt (Date)

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5. API Documentation

Base URL: https://pharmasys-backend.pxxl.click/api

All protected endpoints require a valid JWT token in the Authorization: Bearer <token> header.

Authentication

Method	Endpoint	Description
POST	/auth/login	Login and receive token
GET	/auth/me	Get current user details

Users (Admin only)

Method	Endpoint	Description
GET	/users	List all users
POST	/users	Create a new user
PUT	/users/:id	Update a user
DELETE	/users/:id	Delete a user
PUT	/users/changepassword	Change logged-in user's pa

Medicines

Method	Endpoint	Description
GET	/medicines	Get all medicines

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GET	/medicines/:id	Get a single medicine
POST	/medicines	Add a new medicine
PUT	/medicines/:id	Update a medicine
DELETE	/medicines/:id	Delete a medicine

Patients

Method	Endpoint	Description
GET	/patients	Get all patients
GET	/patients/:id	Get a single patient
POST	/patients	Add a new patient
PUT	/patients/:id	Update a patient
DELETE	/patients/:id	Delete a patient

Prescriptions

Method	Endpoint	Description
GET	/prescriptions	Get all prescriptions
GET	/prescriptions/:id	Get a single prescription
POST	/prescriptions	Create a new prescriptio
PUT	/prescriptions/:id	Update a prescription
DELETE	/prescriptions/:id	Delete a prescription

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Suppliers

Method	Endpoint	Description
GET	/suppliers	Get all suppliers
GET	/suppliers/:id	Get a single supplier
POST	/suppliers	Add a new supplier
PUT	/suppliers/:id	Update a supplier
DELETE	/suppliers/:id	Delete a supplier

Accounts

Method	Endpoint	Description
GET	/accounts/balance	Get current account
GET	/accounts/transactions	List all transactions
POST	/accounts/transactions	Create a new transac
PUT	/accounts/transactions/:id	Update a transaction
DELETE	/accounts/transactions/:id	Delete a transaction

Reports

Method	Endpoint	Description
GET	/reports/sales	Generate sales report
GET	/reports/inventory	Generate inventory repor
GET	/reports/financial	Generate financial report

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Dashboard

Method	Endpoint	Description
GET	/dashboard/stats	Get aggregated stats for dashboa

Settings

Method	Endpoint	Description
GET	/settings	Get pharmacy settings
PUT	/settings	Update pharmacy settings (admin only)

Notifications

Method	Endpoint	Description
GET	/notifications	List all notifications for the
PUT	/notifications/:id/read	Mark a notification as read

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6. Security Considerations

- **Authentication:** JWT tokens with expiration; passwords hashed with bcrypt.
- **Authorization:** Role-based middleware protects routes (e.g., admin-only endpoints).
- **CORS:** Configured to allow only trusted origins (localhost, live domain).
- **Input Validation:** Request data validated using express-validator and Mongoose schemas.
- **Environment Variables:** Sensitive keys (JWT secret, database URI) stored in .env, never committed.
- **Helmet.js:** (Optional) Used to set secure HTTP headers.
- **Data Encryption:** All traffic over HTTPS in production.

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7. Deployment Guide

Backend

1. Set NODE_ENV=production and configure environment variables on the server.
2. Use a process manager like **PM2**: pm2 start server.js --name pharmasys-backend
3. Set up a reverse proxy (e.g., Nginx) to forward requests to port 5000 and enable SSL.

Frontend

1. Build the React app: npm run build (ensure REACT_APP_API_URL points to the live backend).
2. Upload the contents of the build/ folder to your web server's document root.
3. Configure the server to serve index.html for all routes (for client-side routing). Example Nginx:

```
location / {
    try_files $uri /index.html;
}
```

Both services are currently live at the URLs provided.

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8. Future Enhancements

- **Mobile App:** Develop a React Native version for pharmacists on the go.
- **Barcode Scanning:** Integrate barcode scanning for faster checkout and inventory.
- **Insurance Integration:** Direct API connection to insurance providers for claim processing.
- **Automated SMS/Email Reminders:** For prescription refills and appointment reminders.
- **Advanced Analytics:** More detailed sales trends, patient demographics, and forecasting.
- **Multi-branch Support:** Allow a single instance to manage multiple pharmacy locations.

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Conclusion

PharmaSys successfully delivers a modern, reliable, and secure solution for pharmacy management. Its modular architecture, comprehensive feature set, and professional frontend make it suitable for real-world deployment. The system is already live and has been tested with real data, demonstrating its readiness for production use.

We welcome contributions and feedback. For any inquiries, please contact the developer.

— End of Document —

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